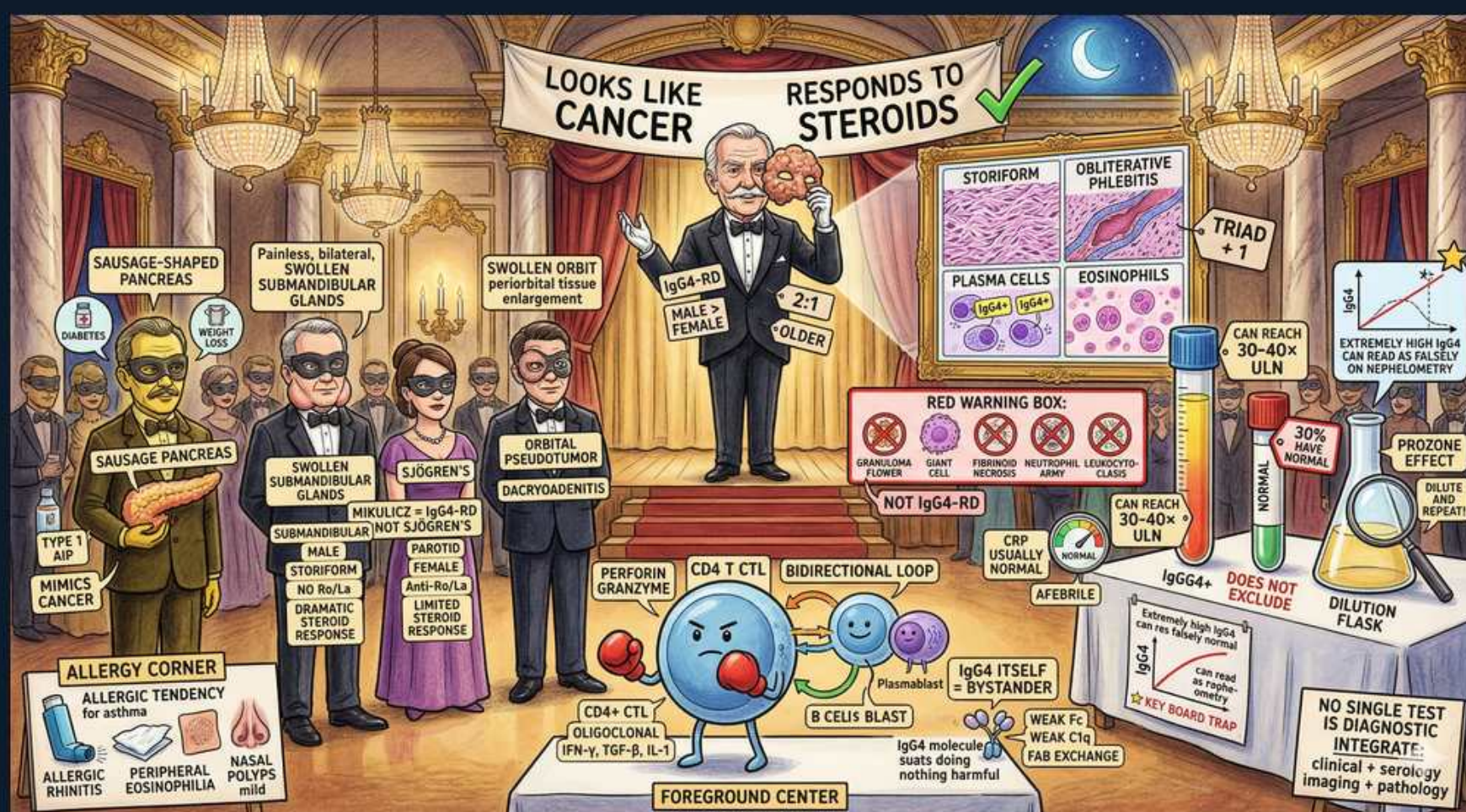


IgG4-Related Disease — 'Looks Like Cancer, Responds to Steroids' Ballroom



1. Looks like cancer / responds to steroids

WHAT YOU SEE

Top banner 'LOOKS LIKE CANCER — RESPONDS TO STEROIDS □'

WHAT IT ENCODES

IgG4-related disease (IgG4-RD) is a systemic autoimmune fibroinflammatory disorder that forms tumor-like masses mimicking malignancy, infection, or other immune diseases, yet shrinks dramatically with corticosteroids — the central teaching point. Pathogenesis involves CD4+ cytotoxic T cells and T-follicular-helper cells driving B-cell/plasmablast expansion, IgG4 class-switching, and progressive fibrosis. The triad of histology is storiform fibrosis + dense lymphoplasmacytic infiltrate rich in IgG4+ plasma cells + obliterative phlebitis. Mnemonic: 'mass that melts with steroids.'

BOARD TRAP

WRONG answer: taking a patient with a pancreatic/orbital/retroperitoneal 'mass' straight to oncologic resection (e.g., Whipple) for presumed cancer. Discriminator: a steroid-responsive mass with elevated serum IgG4 and storiform fibrosis on biopsy is IgG4-RD, not carcinoma — confirm with tissue before irreversible surgery, since the disease regresses on glucocorticoids.

2. Older male predominance

WHAT YOU SEE

Central figure, 'MALE >FEMALE 2:1', OLDER

WHAT IT ENCODES

IgG4-RD characteristically affects older men, with a male-to-female ratio of roughly 2:1 to 4:1 and typical onset in the 50s–70s (middle-aged to elderly). This demographic skew is a useful board discriminator because the closest mimic, Sjögren syndrome, overwhelmingly favors middle-aged women. The age/sex profile, combined with multi-organ mass-forming disease, should push IgG4-RD up the differential.

BOARD TRAP

WRONG answer: diagnosing primary Sjögren syndrome in an older man with bilateral salivary/lacrimal gland swelling. Discriminator: Sjögren favors women, is anti-Ro/La (SSA/SSB) positive with sicca symptoms and focal lymphocytic sialadenitis; IgG4-RD favors older men, is typically anti-Ro/La NEGATIVE, shows IgG4+ plasma cells with storiform fibrosis, and responds briskly to steroids.

3. Storiform fibrosis

WHAT YOU SEE

Histology panel 'STORIFORM' swirling fibrosis

WHAT IT ENCODES

Storiform (cartwheel/whirled, 'straw-mat') fibrosis is the histologic hallmark of IgG4-RD — fibroblasts and inflammatory cells radiate in an irregular swirling pattern. It is one of the two major pathologic features (with obliterative phlebitis) and is required for a histologically confident diagnosis under the consensus criteria, alongside a dense IgG4+ lymphoplasmacytic infiltrate. Late/burnt-out lesions are dominated by this fibrosis with fewer plasma cells, explaining steroid resistance in advanced disease.

BOARD TRAP

WRONG answer: dismissing the biopsy as 'nonspecific chronic inflammation' because only lymphocytes/plasma cells were noted. Discriminator: the specific architecture matters — storiform fibrosis + obliterative phlebitis are the discriminating features that separate IgG4-RD from a generic inflammatory infiltrate; their presence (not just IgG4+ cell counts) anchors the pathologic diagnosis.

4. Obliterative phlebitis

WHAT YOU SEE

Panel 'OBLITERATIVE PHLEBITIS'

WHAT IT ENCODES

Obliterative phlebitis — inflammatory infiltration and partial-to-complete obliteration of vein lumens (while adjacent arteries are spared) — is the second cardinal histologic feature of IgG4-RD. With storiform fibrosis it forms the pathologic backbone of the diagnosis. (Pancreatic and lung lesions show it most reliably; lacrimal/salivary lesions and lymph nodes less so.) An elastic stain helps reveal the obliterated vein.

BOARD TRAP

WRONG answer: concluding IgG4-RD from elevated tissue IgG4+ plasma cell counts ALONE. Discriminator: IgG4+ plasma cells also appear in malignancies (pancreatic cancer), infections, ANCA vasculitis, and other inflammatory conditions; the combination of storiform fibrosis PLUS obliterative phlebitis PLUS an IgG4+/IgG ratio >40% is what makes the histology specific.

5. IgG4+ plasma cells

WHAT YOU SEE

Pink panel 'PLASMA CELLS', IgG4+ cells

WHAT IT ENCODES

A dense lymphoplasmacytic infiltrate rich in IgG4+ plasma cells is the third defining tissue finding. Diagnostic thresholds: an IgG4+/IgG plasma cell ratio >40%, plus an absolute IgG4+ count exceeding an organ-specific cutoff (e.g., >10/hpf in some sites, >50/hpf in pancreas/others). The ratio matters more than the raw count because IgG4+ cells are nonspecific in isolation. Tissue staining, not serum level, is the more reliable diagnostic anchor.

BOARD TRAP

WRONG answer: ruling IgG4-RD in or out on serum IgG4 alone. Discriminator: serum IgG4 is neither sensitive nor specific — it can be normal in active disease and elevated in mimics (pancreatic cancer, Castleman, infection); TISSUE IgG4+ plasma cells with an IgG4+/IgG ratio >40% in the right histologic context is the discriminating finding.

6. Tissue eosinophils

WHAT YOU SEE

Panel 'EOSINOPHILS' in infiltrate

WHAT IT ENCODES

Mild tissue eosinophilia (and sometimes peripheral blood eosinophilia with elevated IgE) frequently accompanies IgG4-RD, reflecting a Th2/allergic component of the immune response. About 30–40% of patients have a personal or family history of atopy. Eosinophils are a supportive, not defining, feature — they help build the picture alongside the obligatory storiform fibrosis and IgG4+ infiltrate.

BOARD TRAP

WRONG answer: labeling marked peripheral eosinophilia + multi-organ disease as eosinophilic granulomatosis with polyangiitis (EGPA/Churg-Strauss). Discriminator: EGPA shows asthma, ANCA (often MPO/p-ANCA) positivity, necrotizing small-vessel vasculitis, and very high eosinophil counts; IgG4-RD shows only MILD eosinophilia, storiform fibrosis, obliterative phlebitis, IgG4+ plasma cells, and is typically ANCA-negative.

7. Sausage-shaped pancreas (Type 1 AIP)

WHAT YOU SEE

Left 'SAUSAGE PANCREAS', painless, mimics cancer

WHAT IT ENCODES

Type 1 autoimmune pancreatitis (AIP) is the prototypic IgG4 manifestation: a diffusely enlarged, 'sausage-shaped' pancreas with a capsule-like rim and irregular narrowing of the main pancreatic duct on imaging. It classically presents with PAINLESS obstructive jaundice, weight loss, and sometimes new-onset diabetes — a near-perfect pancreatic cancer mimic. Serum IgG4 is often markedly elevated; it responds rapidly to steroids.

BOARD TRAP

WRONG answer: reflexively diagnosing pancreatic adenocarcinoma in an older patient with painless jaundice and a pancreatic mass and proceeding to Whipple. Discriminator: a DIFFUSE sausage-shaped gland with duct narrowing, elevated serum IgG4, other-organ involvement, and a dramatic steroid response favors Type 1 AIP; a focal mass with a dilated upstream duct and CA 19-9 elevation favors cancer — get tissue/EUS-FNA before resection.

8. Submandibular swelling / Mikulicz

WHAT YOU SEE

Bilateral swollen submandibular/salivary glands, 'NOT Sjögren'

WHAT IT ENCODES

Painless, firm, bilateral salivary and lacrimal gland enlargement — Mikulicz disease (now recognized as IgG4-RD) and Küttner tumor (chronic sclerosing sialadenitis of the submandibular gland) — is a classic presentation. Unlike Sjögren, sicca symptoms are often mild because glandular fibrosis rather than focal lymphocytic destruction predominates early. It is steroid-responsive.

BOARD TRAP

WRONG answer: calling bilateral gland swelling Sjögren syndrome and ordering only anti-Ro/La. Discriminator: IgG4 Mikulicz disease occurs in older men, is anti-Ro/La NEGATIVE, shows storiform fibrosis with IgG4+ plasma cells on biopsy, and responds to steroids — whereas Sjögren shows positive SSA/SSB, prominent dryness, and focus-score-positive lymphocytic sialadenitis.

9. Orbital pseudotumor / dacryoadenitis

WHAT YOU SEE

Swollen orbit, ORBITAL PSEUDOTUMOR, dacryoadenitis

WHAT IT ENCODES

IgG4-RD is a leading cause of orbital pseudotumor (idiopathic orbital inflammation) and lacrimal gland enlargement (dacryoadenitis), producing painless periorbital swelling, proptosis, and sometimes extraocular muscle involvement or infraorbital nerve enlargement. It responds to corticosteroids and is part of the head-and-neck spectrum with Mikulicz disease.

BOARD TRAP

WRONG answer: assuming a steroid response confirms benign idiopathic orbital inflammation and skipping biopsy. Discriminator: orbital LYMPHOMA (especially MALT lymphoma) can also enlarge lacrimal glands and partially respond to steroids; biopsy with IgG4 staining and flow cytometry/clonality is needed to distinguish IgG4 dacryoadenitis from lymphoma before committing to immunosuppression.

10. Allergy corner

WHAT YOU SEE

'ALLERGY CORNER' — rhinitis, eosinophilia, polyps

WHAT IT ENCODES

A large minority of IgG4-RD patients (around 30–40%) have a background of atopy/allergic disease — allergic rhinitis, asthma, nasal polyps, peripheral eosinophilia, and elevated serum IgE — reflecting the Th2-skewed immune milieu. These features are supportive clues that raise pretest probability, especially when paired with multi-organ mass-forming disease in an older man.

BOARD TRAP

WRONG answer: treating an allergic history plus elevated IgE/eosinophils as sufficient to diagnose IgG4-RD. Discriminator: allergic features are common but nonspecific; the diagnosis still requires the characteristic histology (storiform fibrosis, obliterative phlebitis, IgG4+ plasma cells with IgG4+/IgG >40%) integrated with imaging, organ pattern, and serology — atopy alone proves nothing.

11. Serum IgG4 elevated but not diagnostic

WHAT YOU SEE

Test tube 'IgG4 30–40× ULN', 'DOES NOT EXCLUDE / prozone'

WHAT IT ENCODES

Serum IgG4 is elevated in most but not all patients; roughly 20–40% of biopsy-proven cases have NORMAL serum IgG4, so a normal value does not exclude disease. Conversely, mild elevations are nonspecific (seen in cancer, infection, Castleman disease, vasculitis). Very high IgG4 levels can paradoxically read falsely LOW on nephelometry due to the prozone (antigen-excess) effect — dilution of the sample unmasks the true high value.

BOARD TRAP

WRONG answer: excluding IgG4-RD because serum IgG4 came back normal, or confirming it because the level is mildly high. Discriminator: a normal serum IgG4 does NOT rule out disease (request dilution to defeat prozone if clinical suspicion is high), and an isolated elevation does NOT rule it in — tissue histology remains the diagnostic anchor; serum IgG4 is best for supporting the diagnosis and monitoring treatment response.

12. Often afebrile, normal CRP

WHAT YOU SEE

Red 'warning box': 'AFEBRILE', CRP usually normal

WHAT IT ENCODES

IgG4-RD is typically AFEBRILE with normal or only mildly elevated CRP — it is an indolent fibroinflammatory disease, not an acute febrile/septic one. Disease activity is better tracked by serum IgG4, IgE, eosinophils, plasmablast counts (by flow), and complement (which may fall in active disease) than by CRP. The bland inflammatory markers can falsely reassure clinicians.

BOARD TRAP

WRONG answer: excluding IgG4-RD because the patient is afebrile with a normal CRP. Discriminator: unlike pyogenic infection, Castleman disease, or many vasculitides (which raise CRP/ESR and cause fevers), IgG4-RD usually has bland acute-phase reactants — so a normal CRP and absence of fever should NOT remove it from the differential when imaging and histology fit.

13. Diagnosis = clinical + serology + imaging + pathology

WHAT YOU SEE

Bottom-right 'No single test is diagnostic — integrity'

WHAT IT ENCODES

There is NO single diagnostic test for IgG4-RD; diagnosis integrates the clinical/organ pattern, serum IgG4 (and ancillary IgE, eosinophils, complement, plasmablasts), characteristic imaging, and — most importantly — histopathology showing storiform fibrosis, obliterative phlebitis, and an IgG4+ lymphoplasmacytic infiltrate (IgG4+/IgG >40%). The 2019 ACR/EULAR classification criteria combine inclusion features with exclusion criteria (e.g., must rule out malignancy and mimics).

BOARD TRAP

WRONG answer: making the diagnosis on serology or imaging alone and starting steroids without tissue. Discriminator: because cancer (pancreatic carcinoma, lymphoma) and infection can mimic IgG4-RD clinically, radiologically, and even serologically, biopsy showing the characteristic histology is the load-bearing step — empiric steroids without excluding malignancy can mask and delay a cancer diagnosis.
